
Speaker script

“Back to entropy” — evaluating language models as language models

Deck slides/talk.pdf — 13 slides + 4 backup slides

Target about 8:35 of talking at a relaxed 120 wpm (under 10 minutes), then Q&A

Goal The team leaves with one idea: judge a language model by the loss it was trained on

How to use this document. Italic text is what to say — written to be spoken, not read verbatim. Orange brackets are delivery cues. Times are cumulative. The spoken length is checked by `python tools/script_timing.py`, which fails the build if it passes ten minutes.

The spine — if only this lands, the talk worked:

1. Benchmarks are public and score a model before it is adapted.
2. Loss is the principled yardstick: it is the objective every model was trained on, and in bytes it is comparable across models.
3. Adapt every candidate the same way on fresh, decontaminated text, then measure bits per byte.
4. Adaptation changes the ordering; the new ordering recovers HellaSwag, can be aimed at maths, and predicts which model builds the best fine-tuned system.

SLIDE 1

0:00–0:18

Back to entropy

This talk is about one idea: that we should evaluate language models as language models. I will show why the usual way of choosing a base model misleads, what we do instead, and what we found.

SLIDE 2

0:18–1:11

How good is this model, for what we do?

Every week a new base model ships, and the question is always the same: how good is it, for what we do? A language model is trained to do exactly one thing, predict text. So the direct question is how well it predicts our text. Instead, we read a scorecard. Look at these two small models. On HellaSwag they are two points apart. On our news text, after adaptation, they are three percent apart. On GSM8K they are fifty-five points apart. Two models that model English almost identically well differ by nearly a factor of ten on one exam. Which of these numbers describes the model?

SLIDE 3

1:11–2:02

Two reasons the scorecard misleads

There are two reasons. First, benchmarks are public. They were written before the model existed, vendors optimise for the ones they report, and their content can end up in the training data. A high score might mean a stronger model, or a model that saw more of that exam, and nobody outside can tell which. Second, benchmarks score a model as it ships, but nobody deploys a base model as it ships. We adapt it first. Gemma-4-12B is ninth on GSM8K, eighth on MMLU-Pro, thirteenth of fifteen on raw loss — and once fine-tuned, it builds the best system of all fifteen.

[Land the Gemma example; it comes back on slide 10.]

SLIDE 4

2:02–2:59

Why loss is the right quantity

So what should we measure? There is no universal ground truth for which model is better, so we need a principled yardstick, and loss is the one. Every base model was trained to minimise cross-entropy. Held-out loss is its own objective, measured on text it has not seen. And cross-entropy splits cleanly: the entropy of the text, which is the same for every model, plus the model's distance from the text's true distribution. So a difference in loss is exactly a difference in how close each model is to the text. The only thing stopping a direct comparison is that each model counts in its own tokens. Divide by bytes and that goes away.

[LAND THIS: bits per byte is the training objective on a common scale.]

SLIDE 5

2:59–3:38

Taking that seriously: three requirements

Taking loss seriously asks three things. Bytes, so every tokenizer is on one scale. Fresh, decontaminated text, so no model has trained on it: we use text dated after every model shipped, we deduplicate and split with no overlap, and on the news snapshot we run three membership tests and remove anything any model flags. And equal adaptation: every candidate adapted to the domain by the same procedure, because we want to measure the model we will actually use.

SLIDE 6

3:38–4:34

Equal adaptation: the procedure, and what we ran

The procedure is a per-model Optuna search: a TPE sampler over learning rate, batch, rank and the rest, successive halving to prune weak trials, scored on the validation split, and every model gets the same number of trials. Then the winner trains until validation loss stops improving. That is the default in the released pipeline. Searching seventeen models up to 35 billion parameters was beyond our compute, so the results you will see use the pipeline's other mode: one hand-picked recipe for everyone, 250 steps — minutes per model. And note the trap at the bottom: if you cap the search by time, small models finish more trials and get tuned better.

SLIDE 7

4:34–5:22

Independent views, not ground truth

There is no ground truth for which base model is better. A task tells you which model is better at that task, and different tasks disagree — that is exactly why GSM8K and HellaSwag disagree. So the question is whether the loss agrees with concrete uses it never saw. We built two. One real use: fine-tune fifteen models on a news task. If that were the only task, fine-tuning on it would be the right evaluation, but it needs labels and budget and speaks for one task. And a quiz on articles no model has seen, answered by the original weights, scored leniently and strictly. On news the two views agree at zero point nine two.

SLIDE 8

5:22–6:07

1. Adaptation changes the ordering

First finding. Adaptation does not just lower every score; it reorders the models. After adaptation, seven of eleven models land in exactly their quiz rank. Before, two. Eleven pairs flip from wrong to right and none flip back. The plot shows why: unadapted loss is dominated by cold starts. Gemma opens each block far worse than a model eight times smaller, and has passed it by the end. The unadapted number is measuring familiarity with this kind of prose. A short adaptation removes that, and what is left is capability.

SLIDE 9

6:07–6:47

2. It recovers what benchmarks measure — and can be aimed

Second. A new measurement should recover what existing ones get right. HellaSwag is the benchmark closest to language modelling. Before adaptation our ranking correlates with it at about zero point seven. After adaptation, zero point nine six to zero point nine eight, on three different corpora. And here is the part I find most interesting: adapt on mathematics instead, and the ranking swings toward GSM8K and MMLU-Pro. The choice of text points the measurement at the capability you care about.

SLIDE 10

6:47–7:26

3. It anticipates a real use it never saw

Third, and this is the one that matters. We fine-tuned fifteen models on a news task, and asked how often each way of choosing, fixed before the task existed, agrees with how the fine-tuned systems come out. Adapted bits per byte agrees on ninety-five percent of the pairs. HellaSwag, eighty-seven. GSM8K and MMLU-Pro, a coin flip. Every model lands within one rank of where it finishes — including Gemma, the model the scorecard would never have shortlisted.

[LAND THIS: 0.949 against a coin flip for the reasoning benchmarks.]

SLIDE 11

7:26–8:01

4. It holds

Fourth, it holds up. On the hardest choices, models within two times of each other in size, it agrees with the quiz on ninety percent of pairs, and under lenient scoring it beats simply picking the bigger model. Across four corpora — news, Reddit, Hacker News and maths — it never ranks below a public benchmark. The lead over HellaSwag is consistent but not yet statistically settled; a larger cohort is the next step.

SLIDE 12

8:01–8:27

Using it

In practice: if candidates are more than about two times apart in size, the bigger one usually wins. Otherwise, gather recent text from your domain, adapt every finalist the same way, and rank by adapted bits per byte. Treat anything under two percent as a tie. No labels, no answer key.

SLIDE 13

8:27–8:34

Evaluate language models as language models

So: evaluate language models as language models. Thank you — happy to take questions.

Q&A preparation

“If the fine-tune is the real test, why not just fine-tune?” When you have the task, its labels and the budget, do — it is the better evaluation for that task. A base model is usually chosen before any of those exist and has to serve many tasks. The fine-tune is not ground truth; it is one use the loss was asked to anticipate, and it did.

“Isn’t the quiz circular — built from the same text?” It is answered by the released weights, not the adapted ones, and scored by generation, not likelihood. And the fine-tune, scored on separate documents

against human-written references, agrees with it at Spearman 0.921.

“Doesn’t one hand-picked recipe favour some architectures?” It suits none perfectly. The full method searches each model; we could not afford it at 35B. A second, very different fine-tune recipe left the system ordering at Spearman 0.982.

“Does it beat HellaSwag?” It has the top point estimate everywhere and never falls below it, but at seventeen models the gap is not statistically resolved. We say so in the paper.

“What about contamination of our own text?” Three layers: dated after every release, deduplicated with no split overlap, and membership screens with a quarantine of anything any model flags.

“What does it cost?” With the hand-picked recipe, 3.7 GPU-minutes at 0.5B up to about an hour at 31B on one A100. The corpus is the main cost.